National Institute of Technology Calicut
Department of Computer Science and Engineering

Blockchain Technology Examination

Maximum Marks: 25	**Time Duration:** 1 hour
Date: September 19, 2025	**Academic Year:** 2025-2026

Instructions:

- Attempt all questions.
- Draw neat diagrams wherever required.

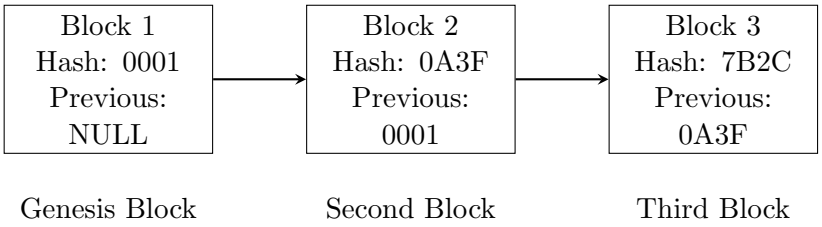
Question 1

Multiple Choice Questions: **(4 marks)**

1. Which cryptographic concept is fundamental to blockchain technology? A. Symmetric encryption B. Hash functions C. Steganography D. Quantum cryptography
2. What is the primary purpose of a consensus mechanism in blockchain? A. To increase transaction speed B. To validate and agree on transactions C. To create new cryptocurrencies D. To reduce energy consumption

Question 2

Study the following blockchain structure diagram and answer the questions below: **(5 marks)**



1. What is the purpose of the "Previous" field in each block?
2. If an attacker tries to modify data in Block 1, what would happen to the hashes in subsequent blocks?

Question 3

The Proof of Work consensus algorithm can be represented by the equation: **(5 marks)**

$$H(nonce||block\ data) < target$$

Explain what each component of this equation represents and how it ensures blockchain security.

Question 4

Based on the following table comparing public and private blockchains: **(5 marks)**

Feature	Public Blockchain	Private Blockchain
Access Control	Anyone can join	Permissioned
Transaction Speed	Slower (5-15 TPS)	Faster (100-1000 TPS)
Consensus	PoW/PoS	Voting-based
Use Cases	Cryptocurrency	Supply chain

Answer the following:

1. Which type of blockchain would be more suitable for a banking system? Why?
2. What is the main trade-off between transaction speed and decentralization?

Question 5

In blockchain networks, transaction verification can be represented using matrices: **(6 marks)**

$$V = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

A value of 1 indicates verification by that node. If the consensus rule requires at least 2 verifications per transaction, which transactions would be accepted? Show your calculations.

END OF QUESTION PAPER